



CSE437: DATA SCIENCE

Group: 07

Section: 01

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1. Introduction

This project uses machine learning models to predict airline passenger satisfaction based on various features such as customer demographics, travel details, and service ratings. In the competitive airline industry, understanding what drives passenger satisfaction is crucial for improving customer experience and maintaining loyalty. The motivation behind this project is to apply machine learning techniques to build a predictive model that can help airlines identify key factors influencing satisfaction and take proactive measures to enhance service quality.

The core problem is accurately classifying passengers as either satisfied or neutral/dissatisfied using models like K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest, and Decision Tree. By leveraging data-driven insights, this project aims to support customer-focused decision-making and improve overall operational strategies within the airline industry.

2. Dataset Description

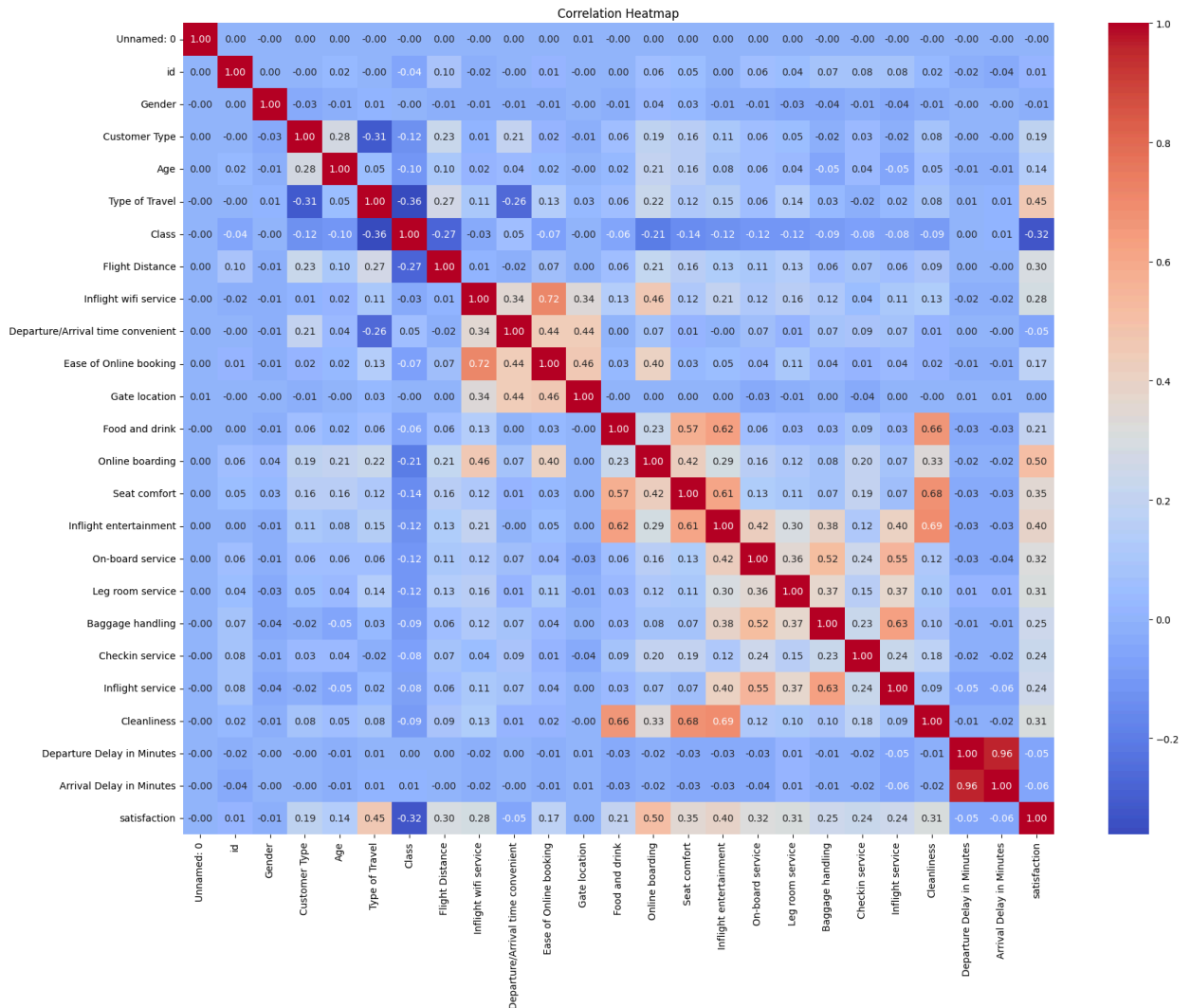
- **Dataset:**  Airline DS
- **Number of Features:** 23 features (both categorical and numerical)

0	Unnamed: 0	103904	non-null	int64
1	id	103904	non-null	int64
2	Gender	103904	non-null	object
3	Customer Type	103904	non-null	object
4	Age	103904	non-null	int64
5	Type of Travel	103904	non-null	object
6	Class	103904	non-null	object
7	Flight Distance	103904	non-null	int64
8	Inflight wifi service	103904	non-null	int64
9	Departure/Arrival time convenient	103904	non-null	int64
10	Ease of Online booking	103904	non-null	int64
11	Gate location	103904	non-null	int64
12	Food and drink	103904	non-null	int64
13	Online boarding	103904	non-null	int64
14	Seat comfort	103904	non-null	int64
15	Inflight entertainment	103904	non-null	int64
16	On-board service	103904	non-null	int64
17	Leg room service	103904	non-null	int64
18	Baggage handling	103904	non-null	int64
19	Checkin service	103904	non-null	int64
20	Inflight service	103904	non-null	int64
21	Cleanliness	103904	non-null	int64
22	Departure Delay in Minutes	103904	non-null	int64
23	Arrival Delay in Minutes	103594	non-null	float64
24	satisfaction	103904	non-null	object

- **Problem Type: Classification**

This is a classification problem as the output feature Satisfaction consists of two discrete categories: “Satisfied and Neutral” or “Dissatisfied”, rather than continuous numerical values.

- **Number of Data Points:** The dataset includes 129880 data points which splits into training and testing sets for model development and evaluation. Training set: 103904 data points and testing set: 25976 data points
- **Feature Types:**
 - **Quantitative:** Unnamed, id, Age, Flight Distance, Inflight wifi service, Departure/Arrival time convenient, Ease of Online booking, Gate location, Food and drink, Online boarding, Seat comfort, Inflight entertainment, On-board service, Leg room service, Baggage handling, Checkin service, Inflight service, Cleanliness, Departure Delay in Minutes, Arrival Delay in Minutes.
 - **Categorical:** Gender, Customer Type, Type of Travel, Class.
- **Correlation Analysis:**



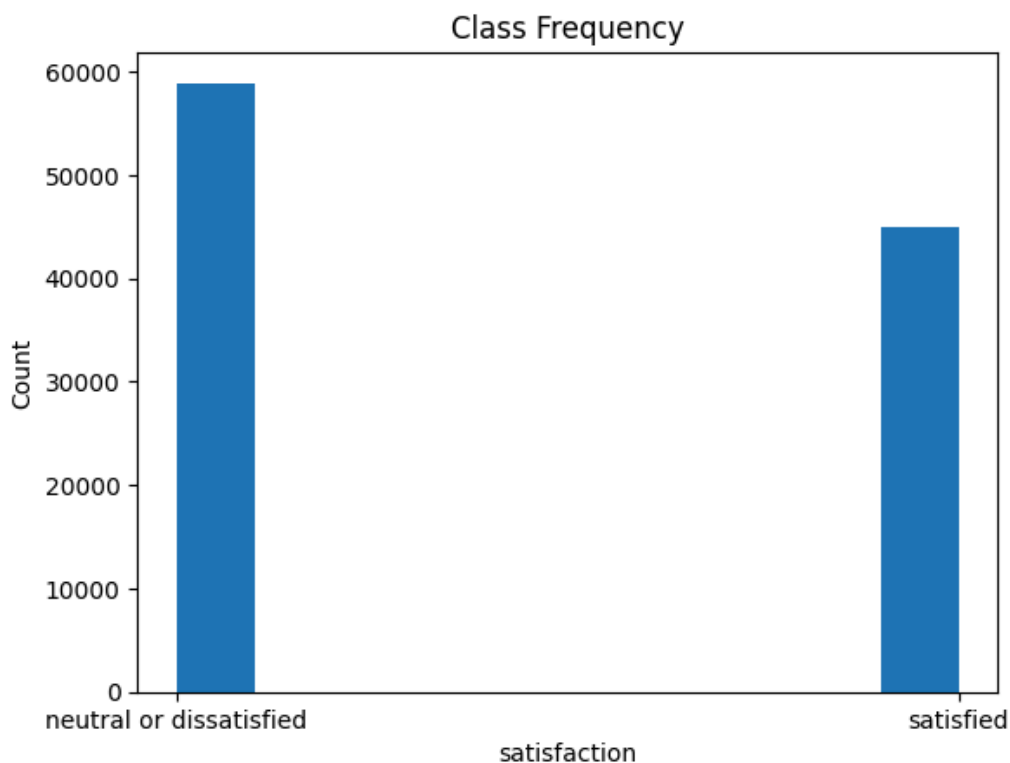
A correlation heatmap was generated using Seaborn to explore the linear relationships between the features and the target variable, Satisfaction. The heatmap revealed that the feature most positively correlated with satisfaction is Online boarding, followed by Type of Travel, and Inflight entertainment. These variables appear to have the strongest linear associations with customer satisfaction.

Features like Gate location, Ease of Online booking, Gender, Departure time and Arrival time show very weak or no correlation with satisfaction, indicating limited linear influence on the target.

Moreover, no features in the dataset show extreme correlation values, above 0.7 or below -0.7, indicating that the features are relatively independent. Given that some features may still hold non-linear relationships with satisfaction, no features were dropped based solely on correlation analysis. All were retained for further exploration

3. Class Distribution and Imbalance

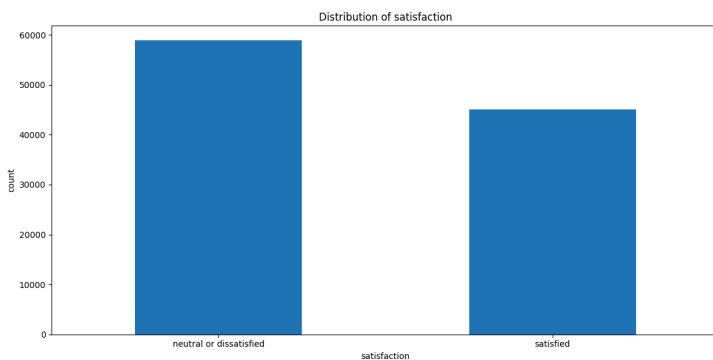
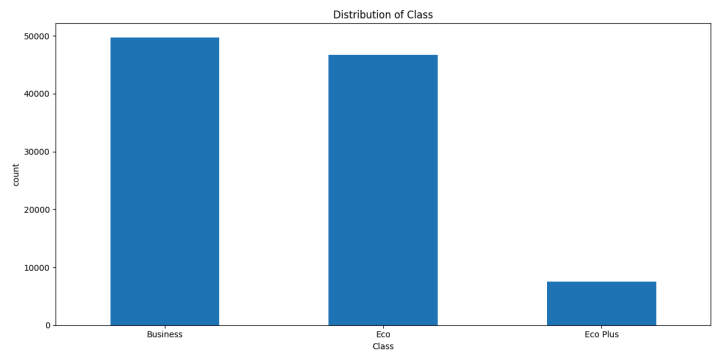
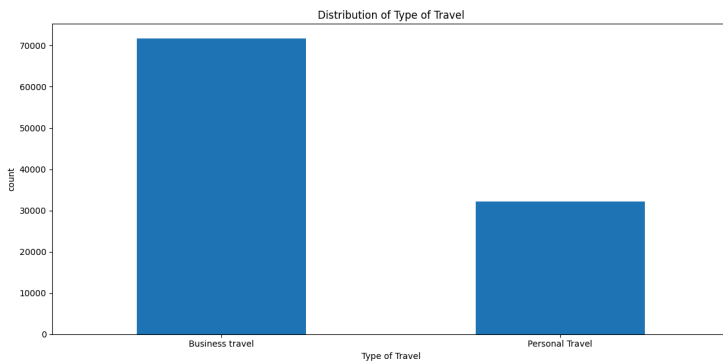
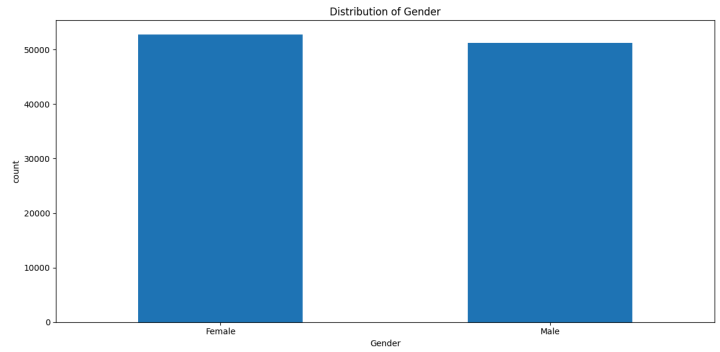
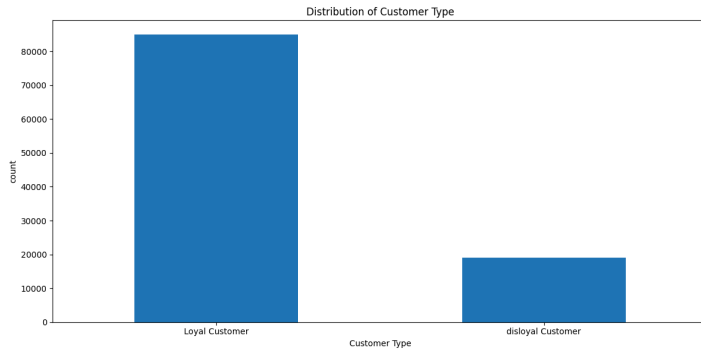
- **Target Feature:** Satisfaction
- **Class Balance:** The classes were visualized using a histogram.



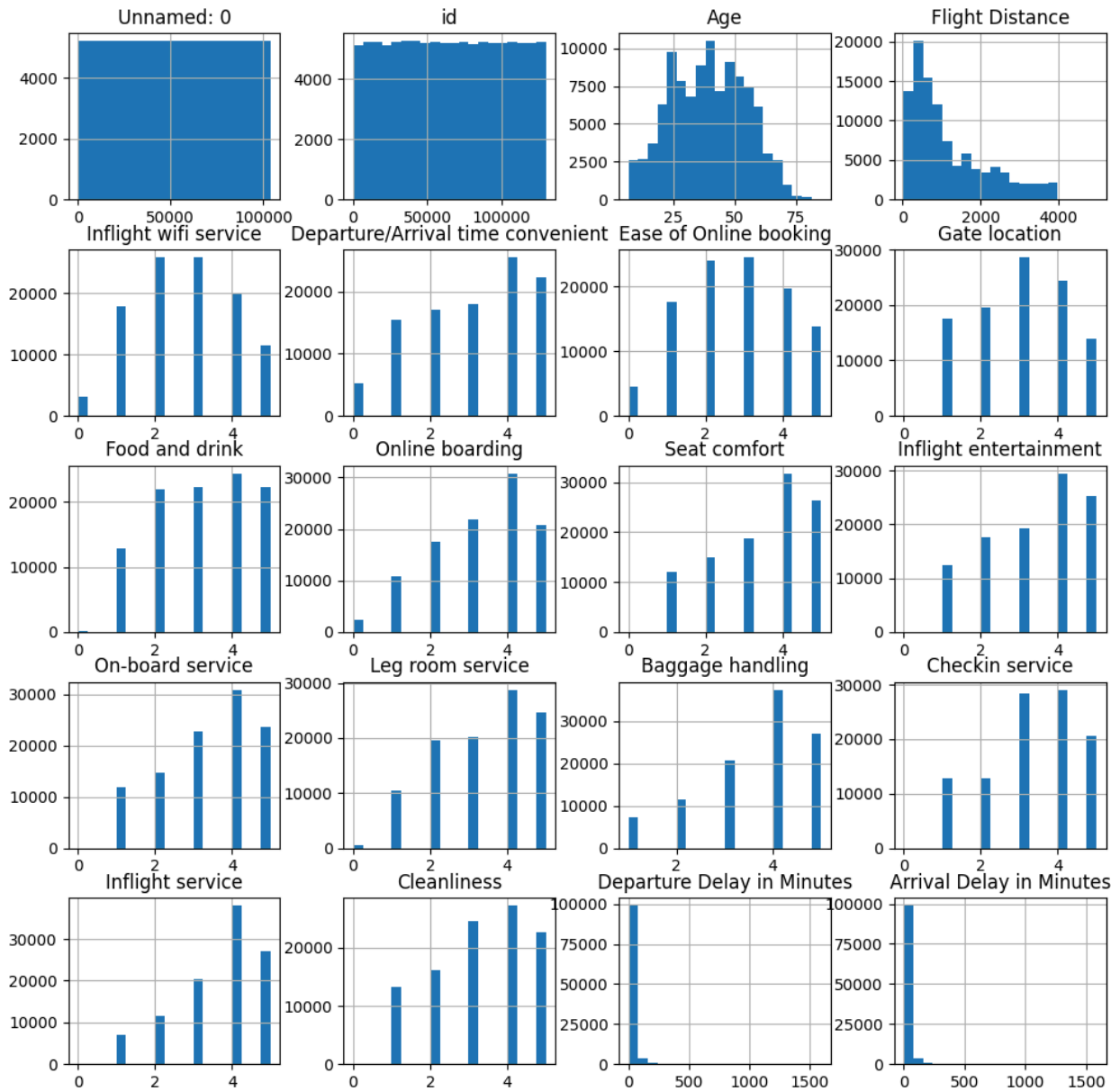
The dataset appears to be slightly imbalanced, with Neutral or dissatisfied having more instances than Satisfied. This imbalance is likely to affect model performance.

4. Exploratory Data Analysis (EDA)

- Target variable, satisfaction, distribution was visualized using a histogram.
- Categorical features were analyzed using bar plots.



- Numerical features were explored using histograms.



- A correlation heatmap was created to examine relationships between numerical features and satisfaction.

5. Dataset Pre-processing

Problem 1: Missing Values

- In the dataset, features like, Arrival Delay in Minutes, contained missing/ null values.
- **Solution:** Since only 1 feature had null values, it was assumed that the null values were considered as 0.

Problem 2: Categorical Encoding

- Machine learning models cannot interpret categorical text data directly.
- **Solution:** Categorical columns were encoded using Label Encoding . In Label Encoding, each category of a feature is mapped to a unique integer. For example, Gender: Male → 0, Female → 1.

Problem 3: Feature Scaling

- Several models used in the project like KNN and SVM are sensitive to the scale of input features.
- **Solution:** Standardization (Normalization) was performed using StandardScaler to numerical features for ensuring consistent scaling. This prevents certain features from dominating due to larger ranges.

Problem 4: Irrelevant Features

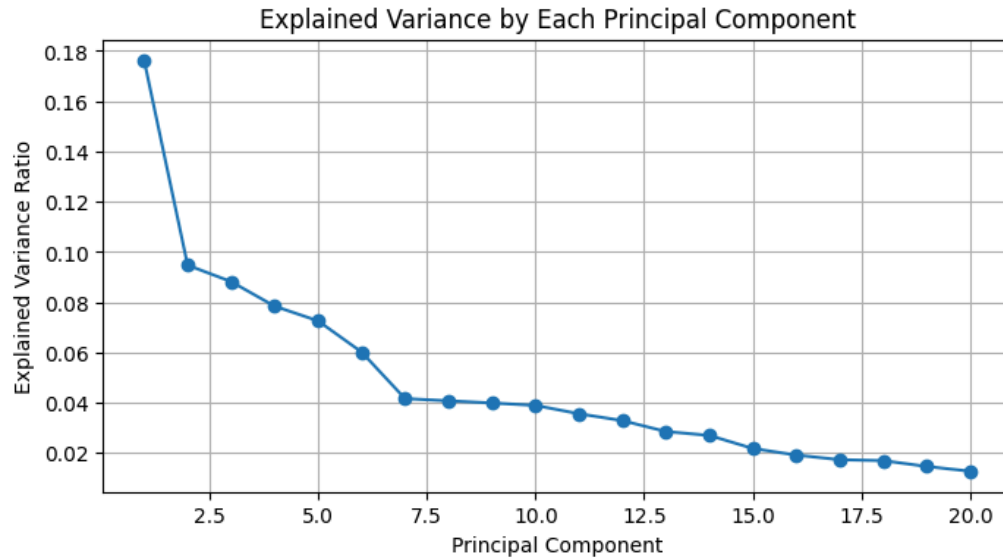
- Features like Gate location and Ease of Online booking showed very weak correlation with Satisfaction and provided little predictive power..
- **Solution:** Based on correlation analysis and PCA loadings, low-impact features will be dropped to improve model performance.

6. Principal Component Analysis

Principal Component Analysis, PCA, was used to reduce dimensionality and identify the most influential features driving variance in the dataset. 20 principal components in total were analyzed. Each component's top 5 contributing features were extracted, and their frequency of appearance was counted to assess overall importance.

- Most frequently appearing feature: Gate location (appeared 10 times)
- Least frequent features: Ease of Online booking, Departure Delay in Minutes, and Arrival Delay in Minutes (1 time each)

Features that appear frequently across top PCs carry significant variance and should be considered important. Low-frequency features that have low correlation or low model importance can be considered for removal .



7. Inferential Statistics

- T-test

8. Dataset Splitting

- **Split Method:** Manual Train-Test Split
- **Training Set:** 80%
- **Testing Set:** 20%

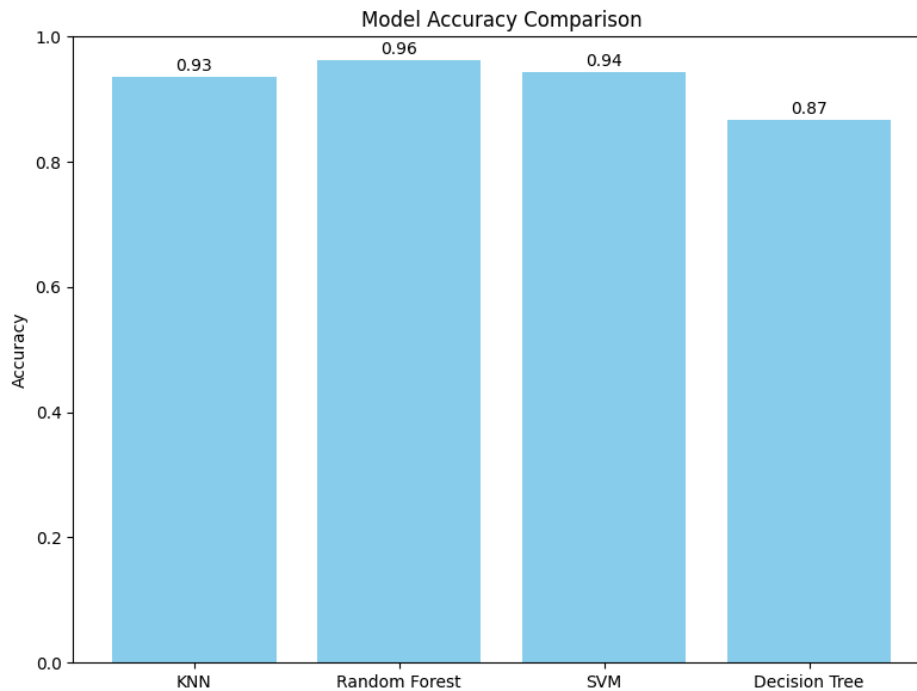
9. Model Training & Testing

In this project, several classification models were trained and evaluated.

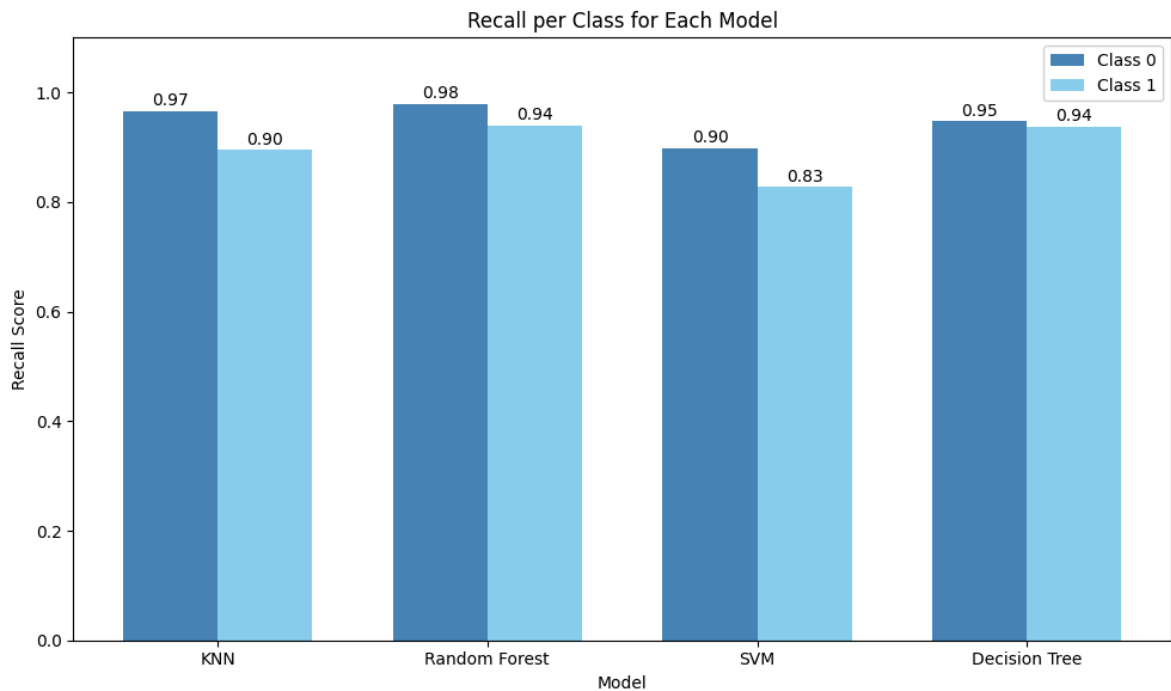
- **K-Nearest Neighbors (KNN):** Used to classify passengers based on similarity to nearby points . It performed moderately well but is sensitive to scaling.
- **Decision Tree:** Used to capture non-linear relationships and was easy to interpret. It performed slightly lower than the rest.
- **Random Forest:** This model delivered a strong accuracy, recall and F1 score.
- **SVM:** In this model, Linear Kernel was used due to the dataset size and it's performance was moderate but training was slower than others.

10. Model Selection/Comparison Analysis

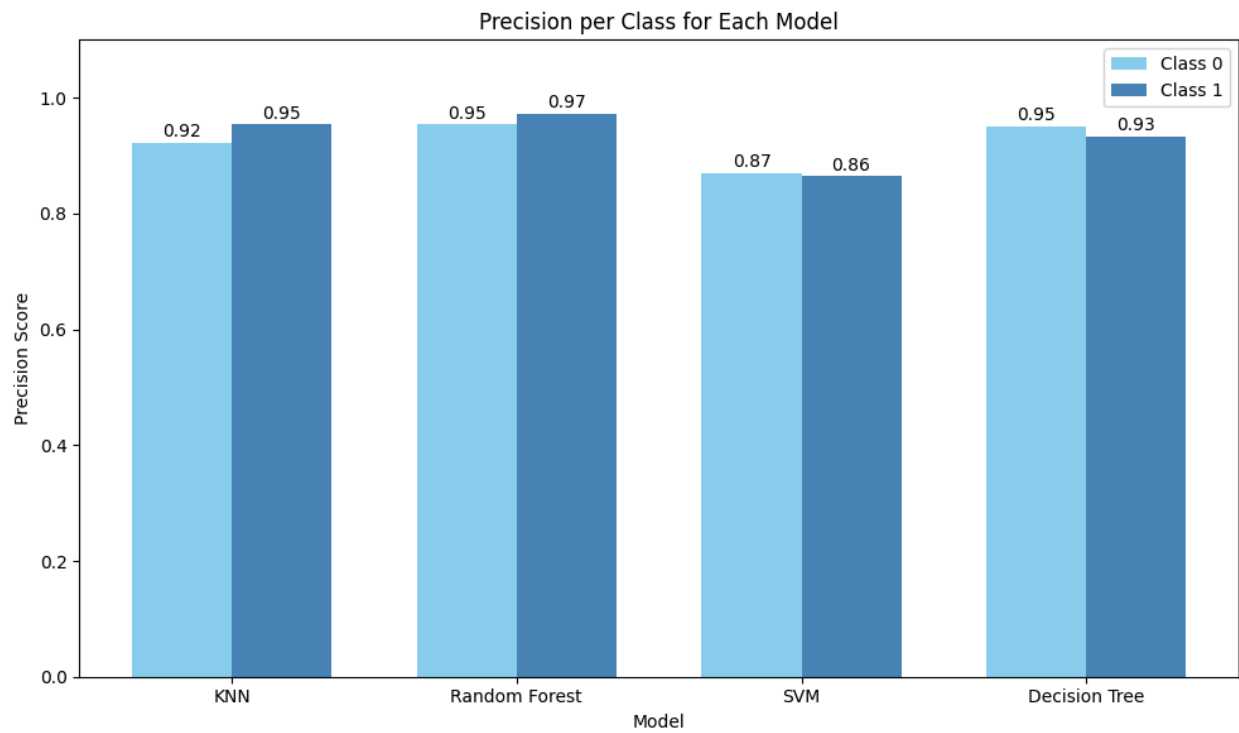
- **Accuracy Bar Chart:** Using matplotlib.pyplot in collab



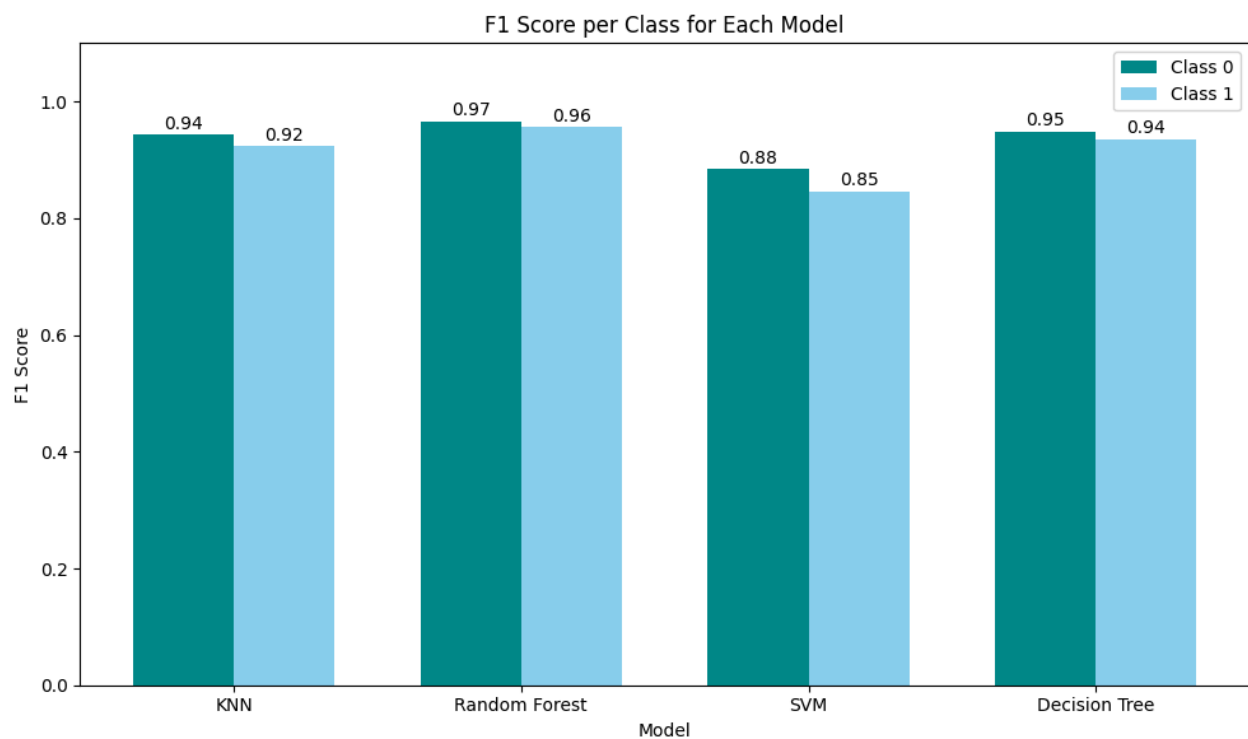
- **Recall Bar Chart:** Using matplotlib.pyplot in collab



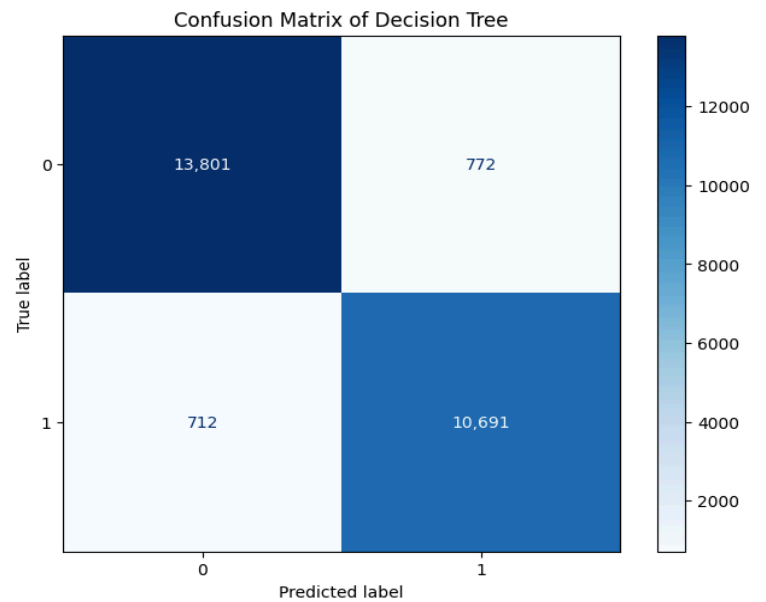
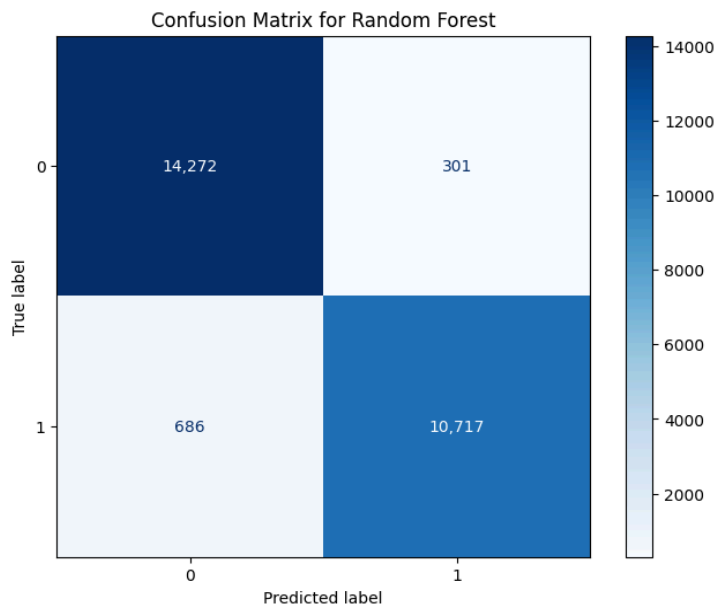
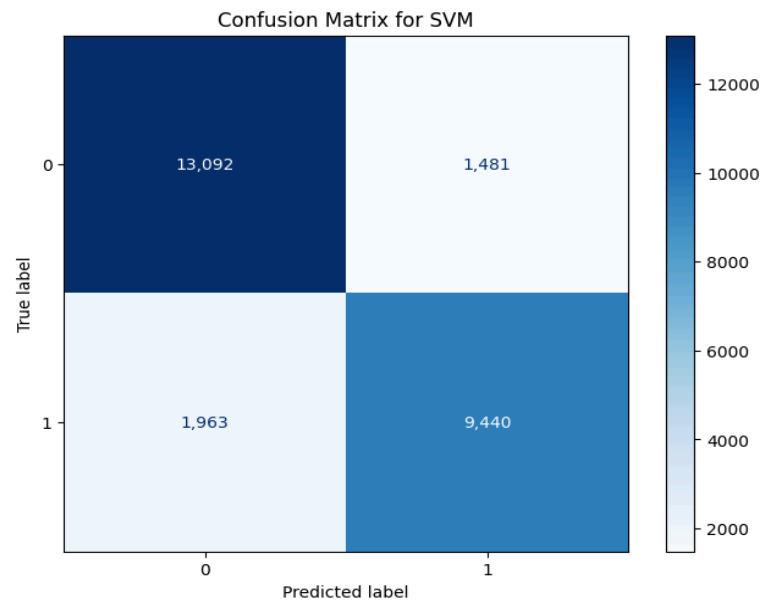
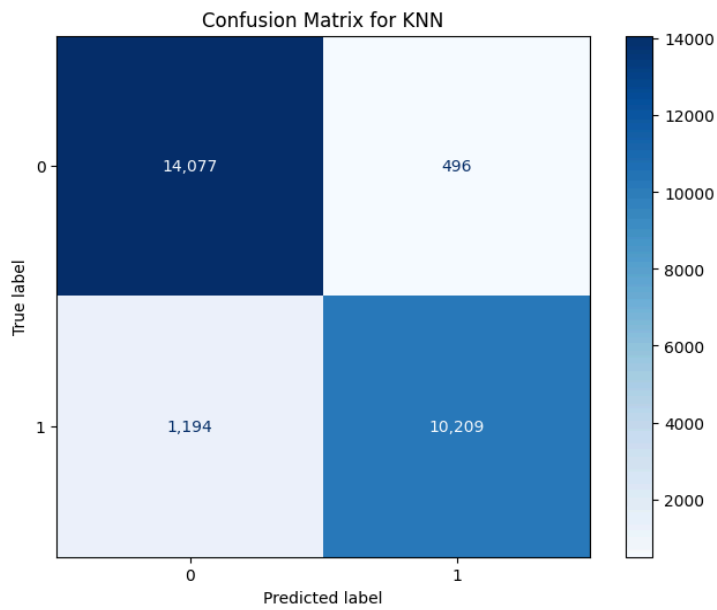
- **Precision Bar Chart:** Using matplotlib.pyplot in collab



- **F1 score Bar Chart:** Using matplotlib.pyplot in collab



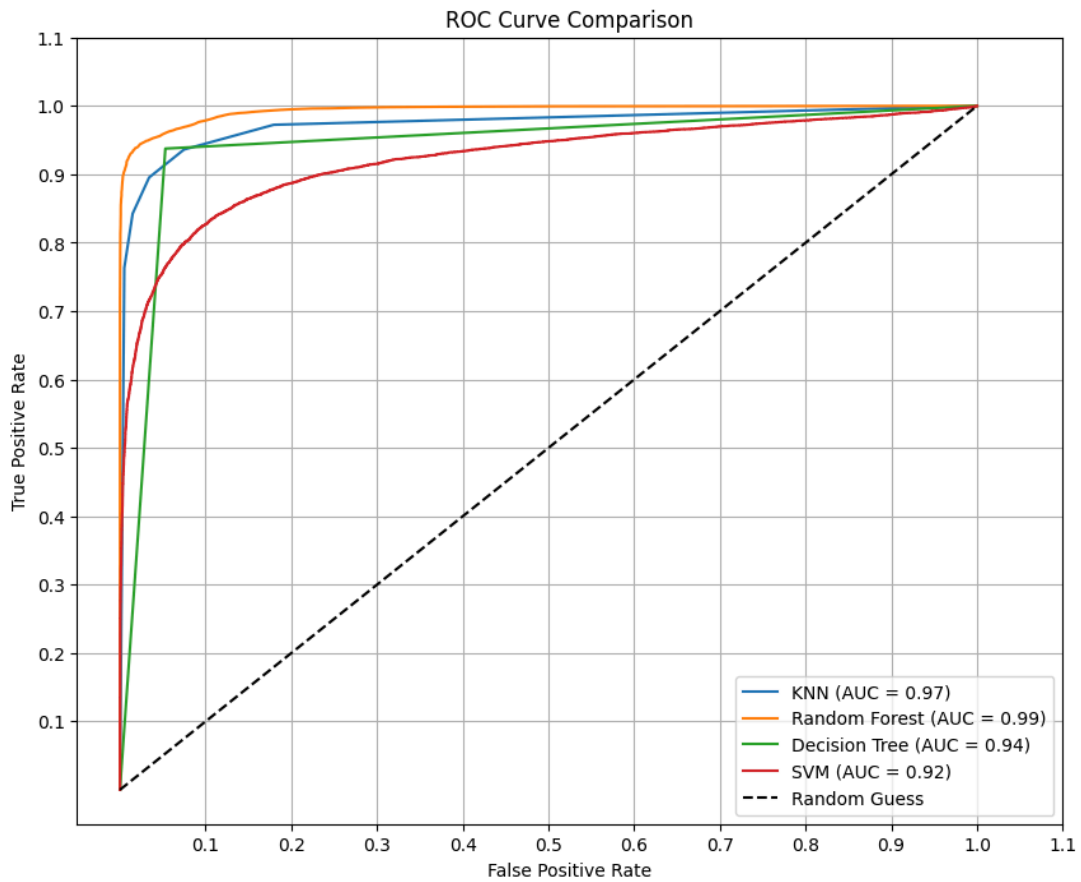
- **Confusion Matrix:** Using matplotlib.pyplot in collab



- **Model Performance Summary:**

Model	Accuracy	Precision	Recall	F1 Score	AUC Score
Random Forest	0.9620	0.9623	0.9620	0.9619	0.99
Decision Tree	0.9429	0.9429	0.9429	0.9429	0.94
KNN	0.9349	0.9358	0.9349	0.9347	0.97
SVM	0.8674	0.8673	0.8674	0.8671	0.92

- **ROC Curve & AUC Score:** Evaluated for binary classification.



- **Summary:**
 - **Random Forest** has the highest accuracy, precision, recall, AUC score and F1 score, making it the best model overall.
 - **Decision Tree** has the second highest accuracy, precision, recall, AUC score and F1 score.
 - **SVM** performed lower overall.

11. Conclusion

This project demonstrated that understanding and predicting passenger satisfaction can be effectively achieved through machine learning. Among the models tested, Random Forest consistently outperformed others in terms of accuracy, precision, recall, F1 score, and AUC score making it the most reliable model for this task. Proper preprocessing of the dataset significantly improved model performance. Overall, the experience highlighted the importance of EDA, feature engineering and smart model selection when working with real-world datasets.